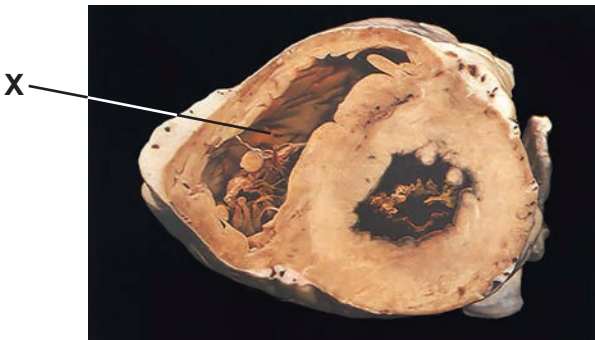


Answer **all** questions.

1. **Image 1.1** shows a transverse section through a mammalian heart.

Image 1.1



(a) (i) Identify the chamber of the heart labelled **X**. [1]

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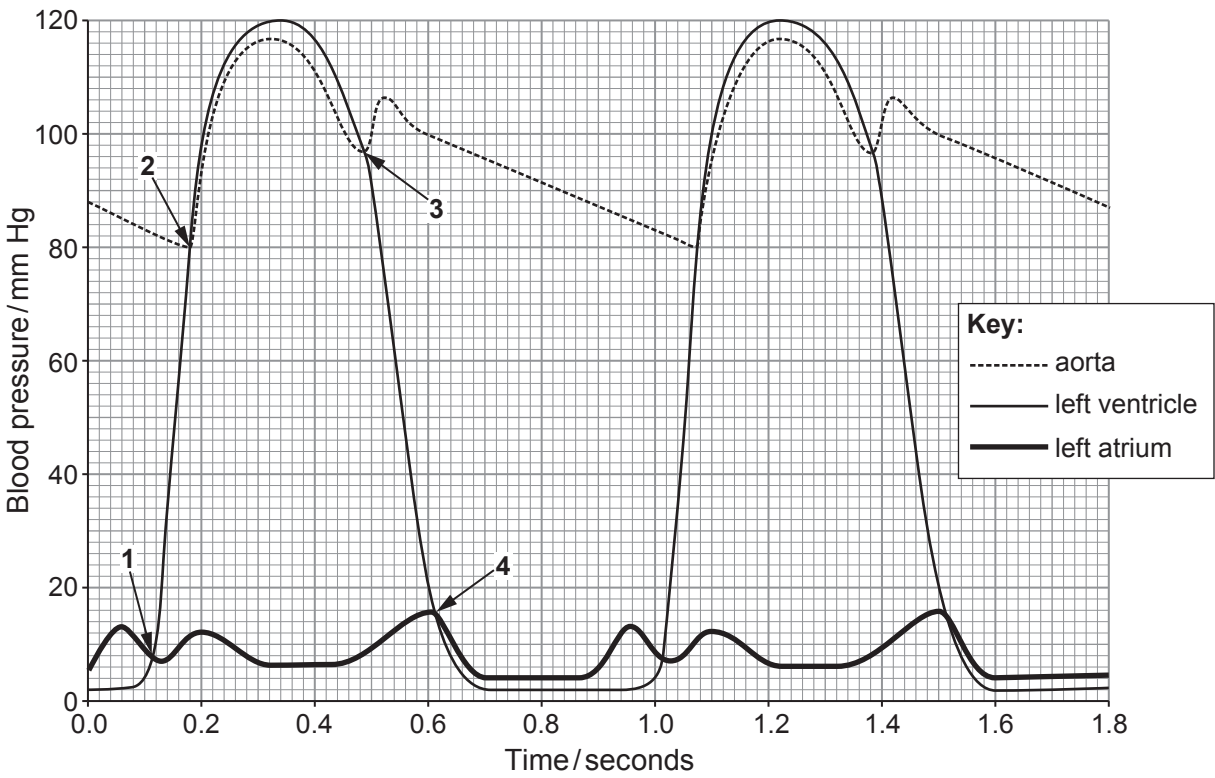
(ii) Explain your answer to (a)(i). [1]

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(b) **Graph 1.2** shows pressure changes in the aorta, left ventricle and left atrium of an adult individual at rest.

Graph 1.2



- (i) On **Graph 1.2 mark with a Y** a point at which blood is flowing from the left ventricle into the aorta. [1]
- (ii) The numbers on **Graph 1.2** indicate the points at which valves in the heart are either opening or closing. Complete the table below to show the name of the valve and state whether the valve is opening or closing. [2]

Number	Name of valve	Opening or closing
1		
3		

- (iii) Calculate the resting heart rate, in beats per minute, for this individual. [1]

Heart rate = beats per minute

- (c) The volume of blood expelled from the left ventricle during each cardiac cycle is known as stroke volume. A person's cardiac output is calculated by multiplying the stroke volume by the heart rate, as shown in the equation below.

Cardiac output = heart rate $\times$ stroke volume

Calculate the stroke volume for an individual whose heart rate was 75 bpm and cardiac output was $5.25\text{ dm}^3\text{ min}^{-1}$. [2]

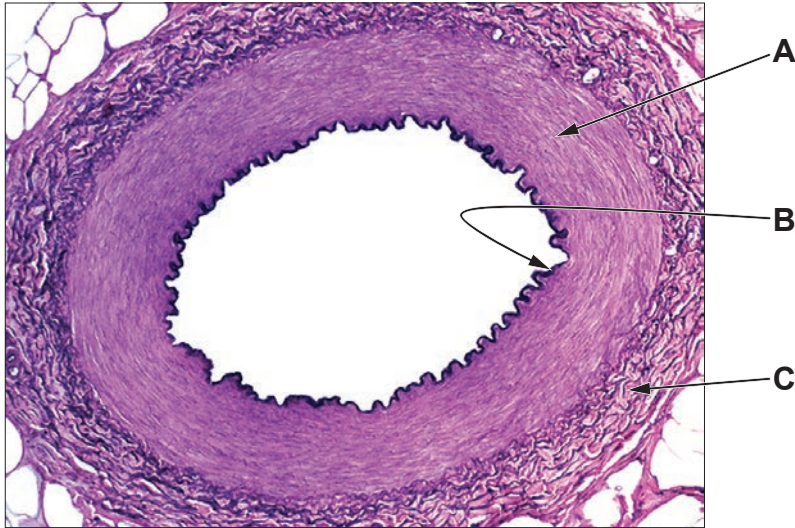
Stroke volume = cm^3

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- (d) **Image 1.3** shows a transverse section through the aorta. The specimen has been stained to provide contrast.

Image 1.3



- (i) Name the tissue layers labelled **A**, **B** and **C**. [2]
- A**
- B**
- C**
- (ii) Although the pressure in the left ventricle drops almost to zero during diastole in the cardiac cycle, the pressure in the aorta remains relatively high. With reference to **Image 1.3**, explain how high pressure is maintained in the aorta. [2]

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